Veterinary Package Insert

TP-COCCILOCK SOL SULFAQUINOXALINE 60MG/ML SOLUTION (6% Sulfaquinoxaline)

PRODUCT DESCRIPTION

TP-Coccilock Sol sulfaquinoxaline 60mg/ml solution brown colour liquid, which contains 6% sulfaquinoxaline, in 5 L bottle.

PHARMACODYNAMICS & PHARMACOKINETICS

Sulfaquinoxaline belongs to a group of antibiotics called the sulphonamides. It is bacteriostatic. Sulfonamides interfere with the biosynthesis of folic acid in bacterial cells; they compete with para-aminobenzoic acid (PABA) for incorporation in the folic acid molecule. By replacing the PABA molecule and preventing the folic acid formation required for DNA synthesis, the sulfonamides prevent multiplication of the bacterial cell. Only organisms that synthesize their own folic acid are susceptible; mammalian cells use preformed folic acid and, therefore, are not susceptible. Cells that produce excess PABA or environments with PABA, such as necrotic tissues, allow for resistance by competition with the sulfonamide.

__Absorption__

Most sulfonamides are well absorbed orally with the exception of the enteric sulfonamides, such as sulfaquinoxaline, which are minimally absorbed. Delays in absorption may occur in adult ruminants or when sulfonamides are administered with food to monogastric animals.

__Distribution__

Sulfonamides are widely distributed throughout the body. They cross the placenta, and a few penetrate into the cerebrospinal fluid. Sulfonamides may be distributed into milk; however, they vary greatly in their ability to do so. The process depends on several factors, including protein binding and pKa values.

__Protein Binding__

Binding can vary depending on serum concentration and other factors.

__Biotransformation__

Sulfonamides are primarily metabolized in the liver but metabolism also occurs in other tissues. Biotransformation occurs mainly by acetylation, glucuronide conjugation, and aromatic hydroxylation in many species. The types of metabolites formed and the amount of each varies depending on the specific sulfonamide administered; the species, age, diet, and environment of the animal; the presence of disease; and, with the exception of pigs and ruminants, even the sex of the animal. Dogs are considered to be unable to acetylate sulfonamides to any significant degree.

N_4-acetyl metabolites have no antimicrobial activity and hydroxymetabolites have 2.5 to 39.5% of the activity of the parent compound. Metabolites may compete with the parent drug for involvement in folic acid synthesis but have little detrimental effect on the bacterial cell, and so could lower the activity of the remaining parent drug.

__Duration of Action__

The sulfonamides have been loosely categorized according to their plasma concentration versus time profiles:

Intermediate- to long-acting—Sulfadimethoxine, sulfamethazine. Enteric (minimally absorbed)—Sulfaquinoxaline.

Note: Duration of action may be estimated by the length of time target serum concentrations are maintained. Target concentrations are generally based on minimum inhibitory concentrations for each organism. Many sources use 50 mcg sulfonamide per mL (5 mg per decaliter) of blood as the minimum effective concentration for sulfonamides in animals.

__Elimination__

Renal excretion is the primary route of elimination for most nonenteric sulfonamides and it occurs by glomerular filtration of parent drug, tubular excretion of unchanged drug and metabolites, and passive reabsorption of nonionized drug. Alkalization of the urine increases the fraction of the dose that is eliminated in the urine. In general, the metabolites of the parent drug are more quickly eliminated by the kidney than the original sulfonamide is, but the proportions of metabolites formed can vary, depending on many factors.

INDICATION

__Chickens:__

i.) As an aid in the control of outbreaks of coccidiosis caused by Eimeria tenella, E. necatrix, E. acervulina, E. maxima, and E. brunetti.

ii.) As an aid in the control of acute fowl cholera caused by Pasteurella multocida susceptible to sulfaquinoxaline and fowl typhoid caused by Salmonella gallinarum susceptible to sulfaquinoxaline.

Turkeys:

i.) As an aid in the control of outbreaks of coccidiosis caused by Eimeria meleagrimitis and E. adenoeides.

ii.) As an aid in the control of acute fowl cholera caused by Pasteurella multocida susceptible to sulfaquinoxaline and fowl typhoid caused by Salmonella gallinarum susceptible to sulfaquinoxaline.

(Note: Please refer to section Recommended Dosage and Administration for the details of dosage according to the indication.)

RECOMMENDED DOSAGE AND ADMINISTRATION

It is to be administered orally.

Chickens:

i.) Administer 125ml of product in 100L of drinking water for 2 or 3 days, skip 3 days then 78.3ml of product in 100L of drinking water for 2 more days. If bloody droppings appear, repeat treatment at 78.3ml of product in 100L of drinking water for 2 more days.

ii.) Administer 125ml of product in 100L of drinking water for 2 or 3 days.

Turkeys:

i.) Administer 78.3ml of product in 100L of drinking water for 2 days, skip 3 days, give for 2 days, skip 3 days and give for 2 more days. Repeat if necessary.

ii.) Administer 78.3ml of product in 100L of drinking water for 2 or 3 days.

(Note: Please refer to section Indication for the details of indication according to the dosage.)

CONTRAINDICATIONS

This product is contraindicated in birds which have uraemia.

Except under special circumstances, this medication should not be used when the following medical problem exists:

» Hypersensitivity to sulfonamides (animals that have had a previous reaction to sulphonamides may be much more likely to react on subsequent administration) or trimethoprim

Risk-benefit should be considered when the following medical problems exist:

Hepatic function impairment (systemically absorbed sulfonamides are metabolized by the liver; delayed biotransformation may increase the risk of adverse effects)

Renal function impairment (systemically absorbed sulfonamides are renally excreted; delayed elimination could cause accumulation of sulfonamide and metabolites, increasing the risk of adverse effects)

WARNINGS & PRECAUTIONS.

May cause toxic reactions unless the drug is evenly mixed in water at dosages indicated and used according to directions.

Do not use in animals hypersensitive (allergic) to sulfonamides or trimethoprim.

People with hypersensitivities (allergies) to sulfonamides or trimethoprim should not handle the medication.

INTERACTIONS WITH OTHER MEDICATIONS

Drug interactions relating specifically to the use of sulfonamides in animals are rarely reported in veterinary literature.

PREGNANCY AND LACTATION

Sulfonamides cross the placenta in pregnant animals. Some teratogenic effects have been seen when very high doses were given to pregnant mice and rats.

It is not recommended for use in pregnant or nursing animals.

Do not use in birds producing eggs for human consumption.

ADVERSE EFFECTS

Crystallization in the urinary tract

Crystallization of sulfonamides can occur in the kidneys or urine with high doses of sulfonamide or when an animal is dehydrated. Solubility in the urine is dependent on the concentration of drug in the urine, urinary pH (less soluble in an acidic pH), the patient's hydration, and the amount of drug in the acetylated form. Because dogs do not produce acetylated metabolites, they may be less susceptible to this adverse effect.It can be minimized in susceptible animals by maintaining a high urine flow and, if necessary, alkalinizing the urine.

Hemorrhagic syndrome (anorexia, epistaxis, hemoptysis, lethargy, pale mucous membranes, possibly death)

Note: Hemorrhagic syndrome has been reported in chickens and dogs but may occur in other species. It is most often reported with the addition of sulfaquinoxaline to feed for chickens, but

in dogs has been reported to follow administration in the water supply of products labeled for poultry.

Sulfaquinoxaline is a vitamin K antagonist that inhibits vitamin K eposide and vitamin K quinone reductase and causes an effect similar to that of coumarin anticoagulants. Rapid hypoprothrombinemia occurs in dogs, and sulfaquinoxaline may have an additional adverse effect on specific cell types; this may explain why supplementation of chicken feeds with vitamin K has not always prevented the syndrome in chickens. Rapid discontinuation of medication and initiation of therapy with vitamin K1 may reverse the effects.

OVERDOSE & TREATMENT

Toxicities secondary to acute overdose of sulfonamides are not typically reported. Side effects may be more likely to occur with high doses and long-term administration, but are seen at recommended doses as well.

Stop medication and to seek for veterinarian advice if there is sign of overdosage.

WITHDRAWAL PERIOD

Chickens & Turkeys: 10 days before slaughter

MAXIMUM RESIDUAL LIMIT (MRL)

Substance / Drug Definition of residues in which MRL was set	Food	Maximum Residue Limits (MRLs) in food µg/kg
Sulfaquinoxaline	Edible offal, muscle (poultry)	100

STORAGE CONDITION

Store in a dry, cool place and avoid direct sunlight. It should be store at room temperature (30^{o}C or below.)

SHELF LIFE

2 years.

After 1st opening, reconstitution or dilution, use within 24 hours.

PACKING

5 L

THYE PHARMA SDN. BHD.
(manufacturer & product registration holder)

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